DESIGN AND DEVELOPMENT OF CAREERPATH AI

USING AWS CLOUD SERVICES

Prepared in the partial fulfillment of the Summer Internship Program on AWS

AT

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Thank you.

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ABSTRACT

CareerPath AI is a web application that helps students and job seekers improve their resumes and choose the right career path. The main aim of this project is to analyze a user's resume, find missing skills, recommend suitable career options, and provide a personalized learning roadmap.

The application is developed using **React** for the frontend and **Spring Boot** for the backend. **Amazon Cognito** is used for secure user login and registration. **Amazon S3** stores the uploaded resume files, and **Amazon DynamoDB** stores resume details and analysis information. AWS Identity and Access Management (IAM) is used to securely connect the application with AWS services.

The application starts by allowing users to register and log in using Amazon Cognito. After logging in, users can upload their resumes in PDF format. The backend reads the resume and extracts important details such as skills, education, and experience. Based on this information, the system calculates a resume score, finds missing skills, recommends suitable career paths, and creates a learning roadmap to help users improve their skills.

The dashboard shows important information such as resume score, career match percentage, missing skills, total uploaded resumes, and the latest resume details. This helps users easily understand their current profile and track their progress.

CareerPath AI combines modern web development with AWS cloud services to provide a secure, simple, and useful platform for career guidance. This project also gives practical experience in React, Spring Boot, REST APIs, AWS services, and full-stack application development.

TABLE OF CONTENTS

1. INTRODUCTION

Finding the right career is an important step for every student and job seeker. Many students have good technical knowledge, but they do not know whether their resume matches the requirements of their desired career. They also find it difficult to identify the skills they need to improve before applying for internships or jobs.

CareerPath AI is a web application developed to help students and job seekers analyze their resumes and choose a suitable career path. The application extracts important information from the uploaded resume, compares the user's skills with predefined career profiles, identifies missing skills, and provides a personalized learning roadmap. This helps users understand their current skill level and improve their knowledge for better career opportunities.

The application is developed using **React** for the frontend and **Spring Boot** for the backend. **Spring Security** and **JWT Authentication** are used to provide secure communication between the frontend and backend. The application is integrated with **Amazon Web Services (AWS)** to provide secure authentication, cloud storage, database management, and application hosting.

The AWS services used in this project include **Amazon Cognito** for secure user registration and login, **Amazon S3** for storing uploaded resume files, **Amazon DynamoDB** for storing resume metadata, **AWS Identity and Access Management (IAM)** for managing secure permissions, and **Amazon EC2** for hosting the backend application.

The project also uses **Docker** to package the application into containers, making deployment easier and more consistent. **Git** and **GitHub** are used for source code management and version control during development.

CareerPath AI is designed as a secure and user-friendly career guidance platform that combines Full Stack Development, AWS Cloud Services, and containerization to help users improve their resumes and prepare for their careers.

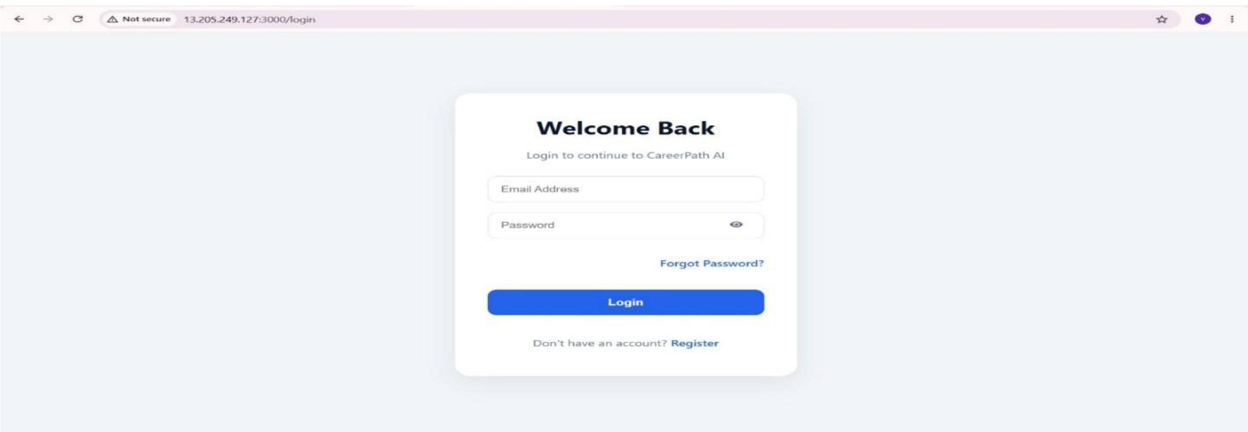

Figure 1.1 CareerPath AI Login Page

Importance of the Project

Many students prepare resumes without knowing whether they have the required skills for a particular career. CareerPath AI helps users understand how well their skills match different career profiles.

After a resume is uploaded, the application extracts important information such as skills, education, projects, and experience. The extracted skills are then compared with **predefined career profiles stored in a JSON file**. Based on the matching skills, the application recommends the most suitable career path, identifies missing skills, and generates a personalized learning roadmap.

The application also provides a dashboard where users can view their resume score, career match percentage, missing skills, uploaded resumes, and resume history.

Amazon Cognito, Amazon S3, Amazon DynamoDB, IAM, Amazon EC2, and Docker make the application secure, reliable, and easy to deploy.

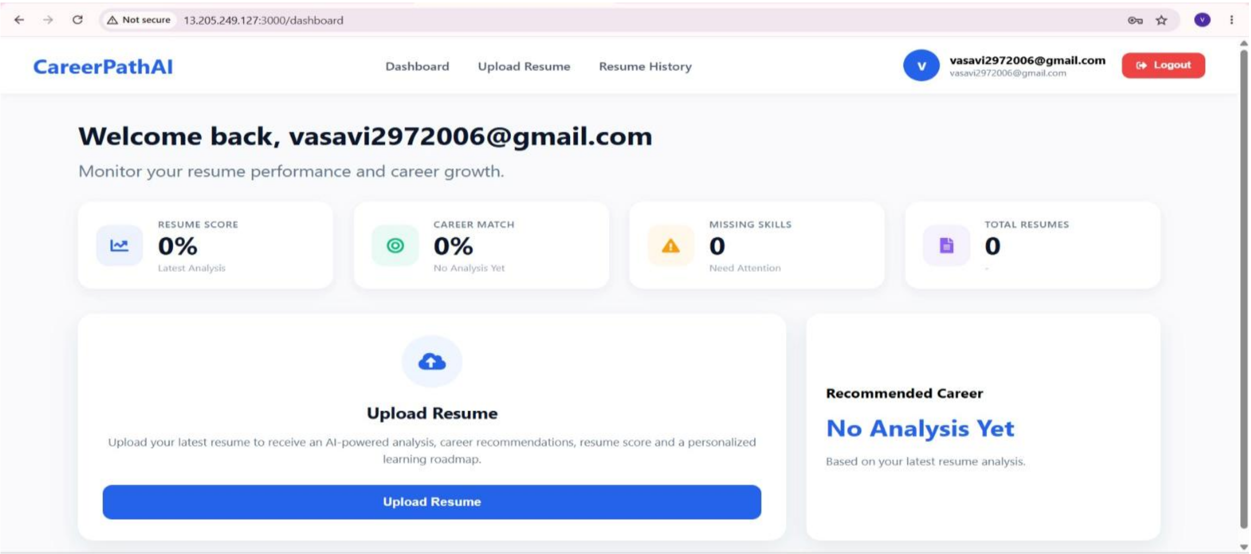

Figure 1.2 Dashboard

Objectives of the Project

The main objectives of CareerPath AI are:

- To build an AI-powered career guidance platform.
- To provide secure user authentication using Amazon Cognito.
- To upload and store resumes securely in Amazon S3.
- To store resume information in Amazon DynamoDB.
- To analyze resumes and calculate a resume score.

- To compare user skills with predefined career profiles stored in a JSON file.
- To identify missing skills and recommend career paths.
- To generate a personalized learning roadmap.
- To containerize the application using Docker.
- To develop a Full Stack application using React, Spring Boot, and AWS.

Scope of the Project

CareerPath AI is mainly designed for students, fresh graduates, and job seekers. It helps users understand how their current skills match different career profiles and provides guidance on improving their knowledge.

Currently, the application uses **predefined career profiles** stored in a JSON file for skill matching. In the future, the project can be enhanced by adding more career profiles, support for additional resume formats, company-specific skill requirements, and improved recommendation techniques.

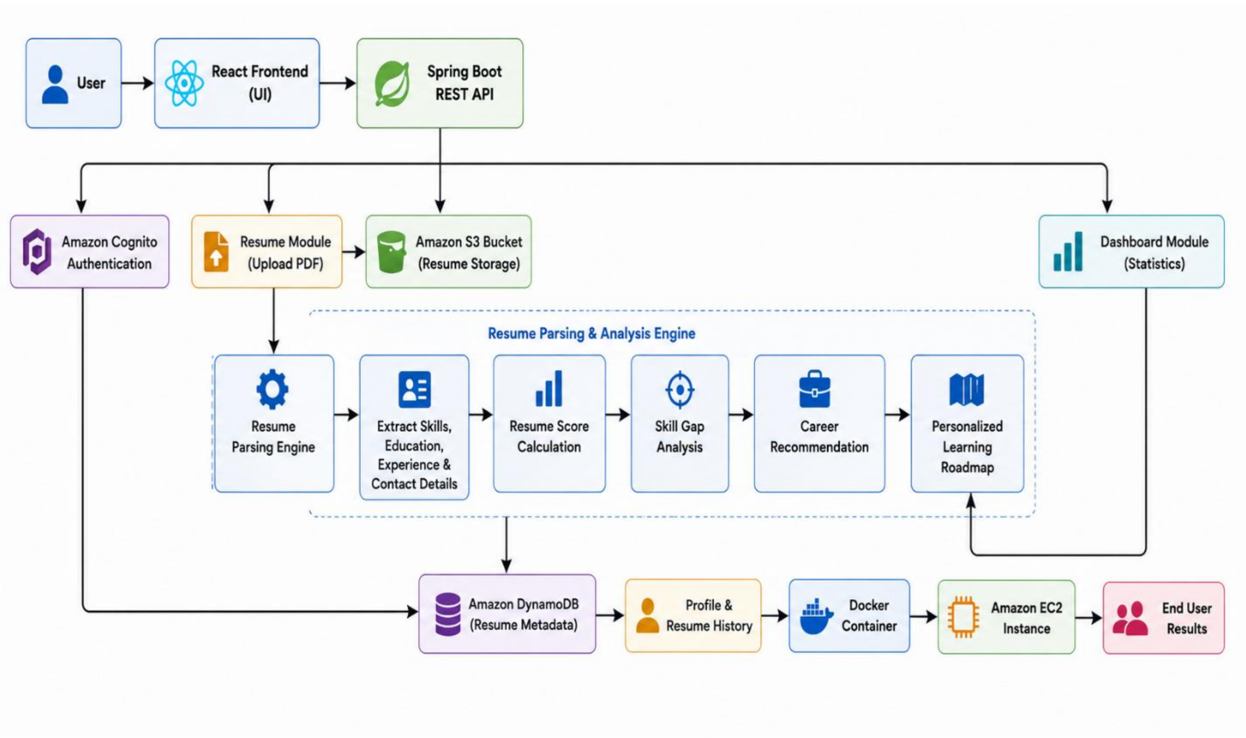

Figure 1.3 CareerPath AI Overview

2. METHODOLOGY

The CareerPath AI project was developed using a step-by-step approach. Each module was implemented separately and then integrated into the complete application. The main goal was to build a secure and user-friendly platform that helps users analyze their resumes and receive career recommendations.

The development process started with understanding the project requirements. After planning the application, the frontend and backend were developed, followed by AWS integration and testing. The methodology used in this project is explained below.

2.1 Requirement Analysis

The first step was to identify the requirements of the application. The system was designed to help users upload resumes, analyze their skills, and receive career guidance.

The main requirements are:

- User Registration and Login
- Resume Upload
- Resume Analysis
- Career Recommendation
- Learning Roadmap
- Dashboard
- Resume History

These requirements were used to design the complete application.

2.2 System Design

After collecting the requirements, the application architecture was designed.

The system consists of three main parts:

- React Frontend
- Spring Boot Backend
- AWS Cloud Services

The frontend provides the user interface.

The backend processes the user requests and performs resume analysis.

AWS services are used for authentication, file storage, database management, and application hosting.

2.3 Application Workflow

The application follows a simple workflow.

First, the user creates an account and logs in using Amazon Cognito. After successful login, the user uploads a resume in PDF format. The resume is stored securely in Amazon S3.

The backend reads the uploaded resume and extracts important information such as skills, education, projects, and experience.

The extracted skills are then compared with **predefined career profiles stored in a JSON file**. Based on the matching skills, the system calculates the career match percentage and identifies missing skills.

Finally, a personalized learning roadmap is generated, and all the results are displayed on the dashboard.

2.4 Security

Security is an important part of the application.

The following security features are implemented:

- Amazon Cognito for Authentication
- JWT Token Validation
- Spring Security
- Secure API Access
- IAM Permissions

These features ensure that only authenticated users can access the application.

2.5 AWS Integration

AWS services are used to improve the security and reliability of the application.

The AWS services used are:

- Amazon Cognito
- Amazon S3
- Amazon DynamoDB
- AWS IAM
- Amazon EC2

Each service performs a specific task within the application.

2.6 Docker Integration

Docker is used to package the backend application into a container.

The Docker container includes all the required dependencies, making the application easy to deploy on different systems without changing the configuration.

2.7 Testing

After development, every module was tested individually.

The following modules were tested:

- User Registration
- Login
- Resume Upload
- Resume Analysis
- Dashboard
- Resume History
- Career Recommendation

The application worked successfully after integrating all the modules.

The methodology explains the steps followed to develop the CareerPath AI application. The project started with requirement analysis, followed by system design, frontend and backend development, AWS integration, testing, and deployment. The application allows users to register, log in, upload resumes, and receive career recommendations. The uploaded resume is processed to extract skills, which are compared with predefined career profiles stored in a JSON file. Based on the matching skills, the system recommends a suitable career path, identifies missing skills, and generates a personalized learning roadmap. Finally, the application was tested to ensure all modules worked correctly.

3. TECHNOLOGY STACK

The CareerPath AI application is developed using modern web technologies and Amazon Web Services (AWS). Each technology is used for a specific purpose to build a secure, scalable, and user-friendly application. React is used to develop the frontend, while Spring Boot is used to build the backend. Spring Security and JWT Authentication provide secure access to the application. AWS services are used for user authentication, cloud storage, database management, and application hosting. Docker is used for containerization, and Git with GitHub is used for source code management.

The technologies used in this project are shown in the table below.

3.1. Technology Stack

Category	**Technology**	**Purpose**
Frontend	React	Develops the user interface
Styling	CSS3	Designs responsive web pages
Backend	Spring Boot	Handles business logic and REST APIs
Programming Language	Java 21	Backend development
Security	Spring Security	Secures application APIs
Authentication	JWT	Secure API communication
Build Tool	Maven	Builds and manages the project
Containerization	Docker	Packages the application
Version Control	Git	Tracks source code changes
Repository	GitHub	Stores project source code
API Testing	Swagger	Tests and documents REST APIs

The combination of these technologies provides a complete Full Stack application with secure authentication, efficient backend processing, and a user-friendly interface.

3.1.1 React

React is used to build the frontend of the application. It provides a fast and interactive user interface where users can register, log in, upload resumes, and view their analysis results.

3.1.2 Spring Boot

Spring Boot is used to develop the backend application. It handles REST APIs, resume processing, career recommendations, and communication with AWS services.

3.1.3 Spring Security

Spring Security protects the backend APIs by allowing only authenticated users to access the application.

3.1.4 JWT Authentication

JWT (JSON Web Token) is used to verify user identity after successful login and securely communicate between the frontend and backend.

3.1.5 Docker

Docker is used to package the backend application into a container. This makes deployment simple and ensures that the application runs consistently in different environments.

3.1.6 Git and GitHub

Git is used to track changes in the source code, while GitHub is used to store and manage the project repository.

3.1.7 Swagger

Swagger provides interactive documentation for REST APIs and allows developers to test backend endpoints easily during development.

3.2. AWS Services Used

3.2.1 Amazon Cognito

Amazon Cognito provides secure user authentication. It manages user registration, login, email verification, forgot password, and reset password features.

3.2.2 Amazon S3

Amazon S3 is used to store uploaded resume files securely. The backend retrieves the files whenever resume analysis is performed.

3.2.3 Amazon DynamoDB

Amazon DynamoDB is used to store resume metadata and analysis information such as resume score, recommended career path, and upload details.

3.2.4 AWS Identity and Access Management (IAM)

IAM manages secure permissions between the application and AWS services. It allows the backend to access Amazon S3 and DynamoDB securely.

3.2.5 Amazon EC2

Amazon EC2 is used to host the Spring Boot backend application. It provides a cloud server that allows users to access the application through the internet.

3.2.6 Elastic IP

Elastic IP provides a fixed public IP address for the EC2 instance. This ensures that the backend URL remains the same even after restarting the EC2 instance.

3.3 Advantages of the Technology Stack

The technologies and AWS services used in this project provide several advantages:

- Easy-to-use and responsive user interface.
- Secure user authentication.
- Reliable cloud storage for resumes.
- Fast and scalable database management.
- Secure API communication.
- Simple application deployment using Docker.
- Reliable cloud hosting using Amazon EC2.
- Efficient source code management with Git and GitHu

4. SYSTEM DESIGN AND ARCHITECTURE

The system architecture shows how different components of the CareerPath AI application work together. The application follows a three-tier architecture consisting of the frontend, backend, and AWS cloud services. The React frontend provides the user interface, while the Spring Boot backend processes user requests and communicates with AWS services. Amazon Cognito is used for user authentication, Amazon S3 stores uploaded resumes, Amazon DynamoDB stores resume information, IAM provides secure access, and Amazon EC2 hosts the backend application. Docker is used to deploy the backend application on the EC2 instance. The following sections explain each component of the system architecture.

4.1 Overall System Architecture

The CareerPath AI application is designed using a three-tier architecture. The frontend is developed using React and provides pages for user registration, login, resume upload, dashboard, and resume history. The backend is developed using Spring Boot, which handles authentication, resume processing, career recommendation, and communication with AWS services. The AWS cloud provides authentication, file storage, database management, and application hosting. This architecture improves security, scalability, and application performance.

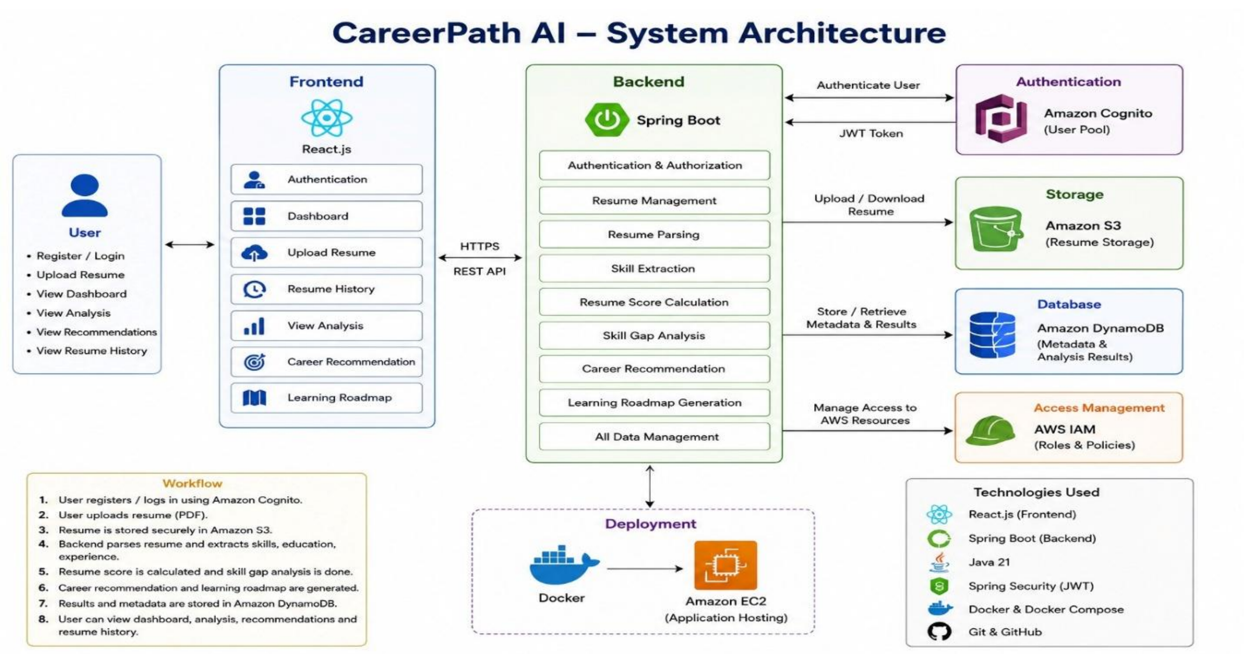

Figure 4.1 Overall System Architecture

4.2 Authentication Layer – Amazon Cognito

Amazon Cognito is used to provide secure user authentication. It allows users to create an account, verify their email address, log in securely, and reset their password if required. After successful login, Amazon Cognito generates authentication tokens that are validated by Spring Security before allowing access to protected APIs. This ensures that only authenticated users can use the application.

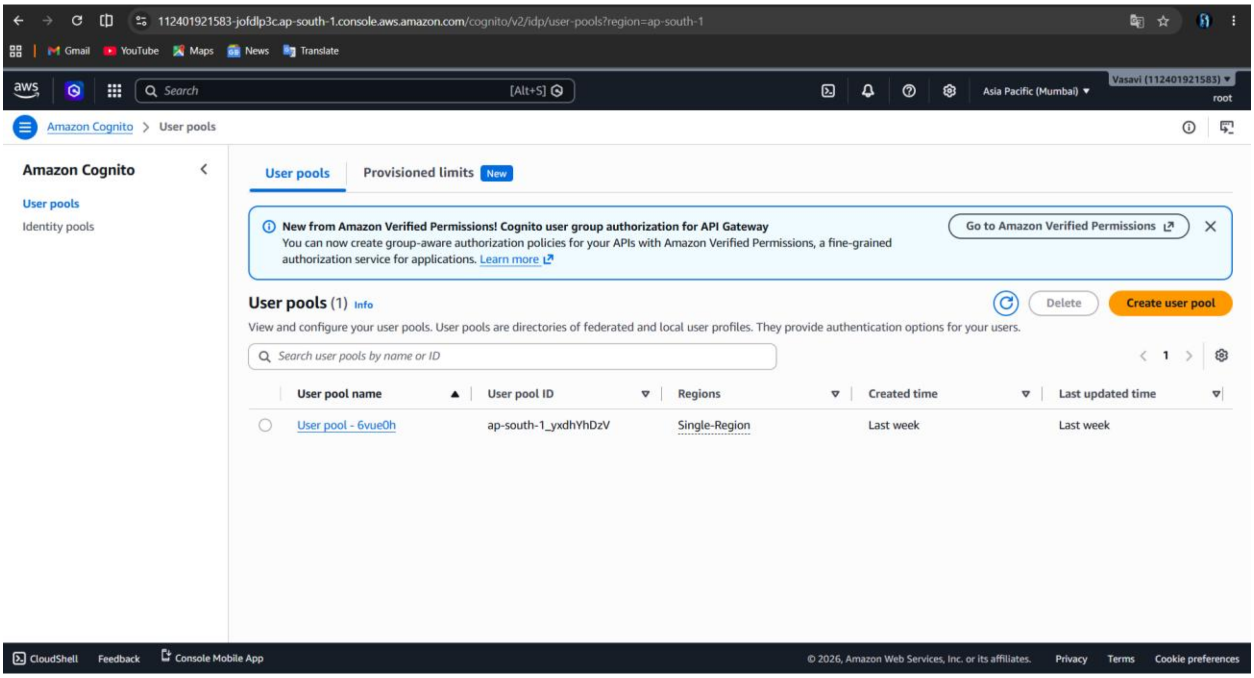

Figure 4.2 Amazon Cognito User Pool

4.3 Resume Storage Layer – Amazon S3

Amazon S3 is used to store uploaded resume files securely. When a user uploads a resume in PDF format, the backend stores the file in an S3 bucket. The stored resume can later be accessed whenever resume analysis is required. Using Amazon S3 provides secure, reliable, and scalable cloud storage for resume files.

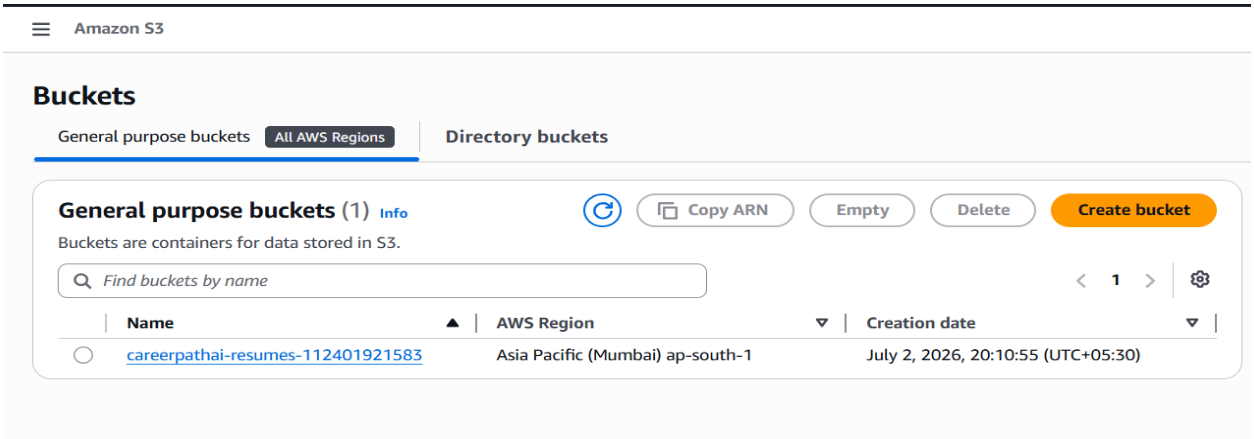

Figure 4.3.1 Amazon S3 Bucket

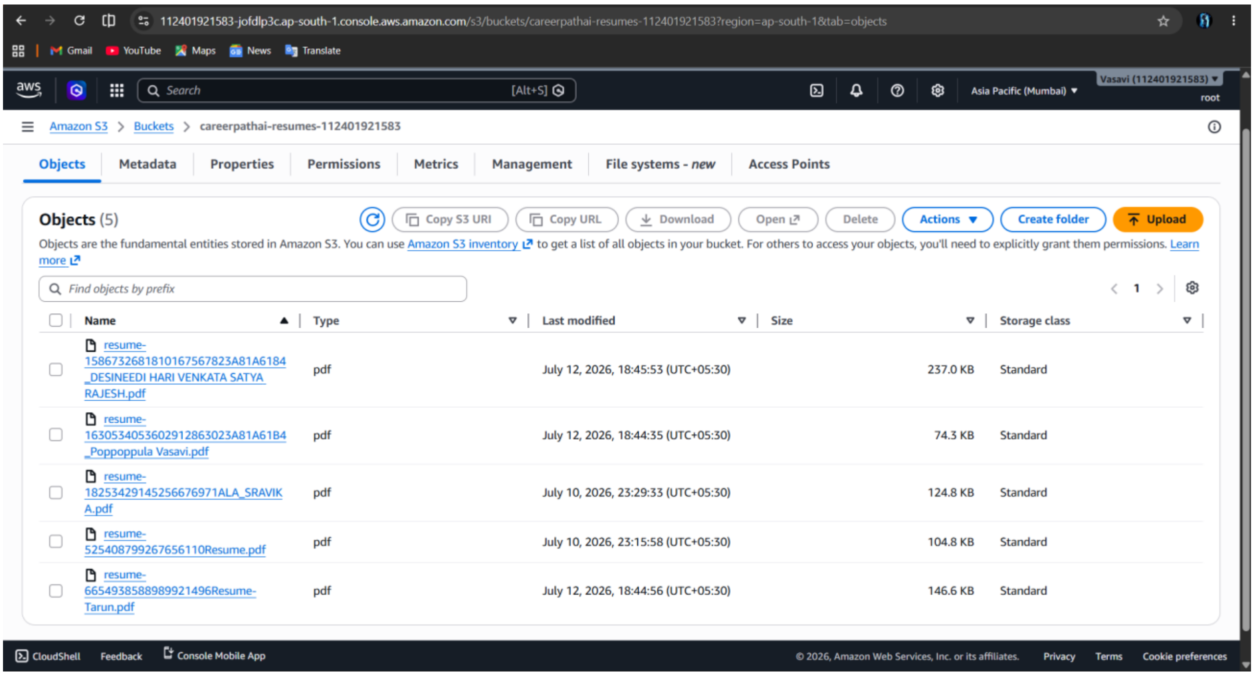

Figure 4.3.2 Resumes Files Uploaded into Amazon S3 Bucket

4.4 Resume Processing Layer

After the resume is uploaded, the backend extracts important information such as skills, education, projects, and experience. The extracted skills are compared with **predefined career profiles stored in a JSON file**. Based on the matching skills, the application recommends the most suitable career path, identifies missing skills, and generates a personalized learning roadmap. This process helps users understand their current skill level and improve the required skills.

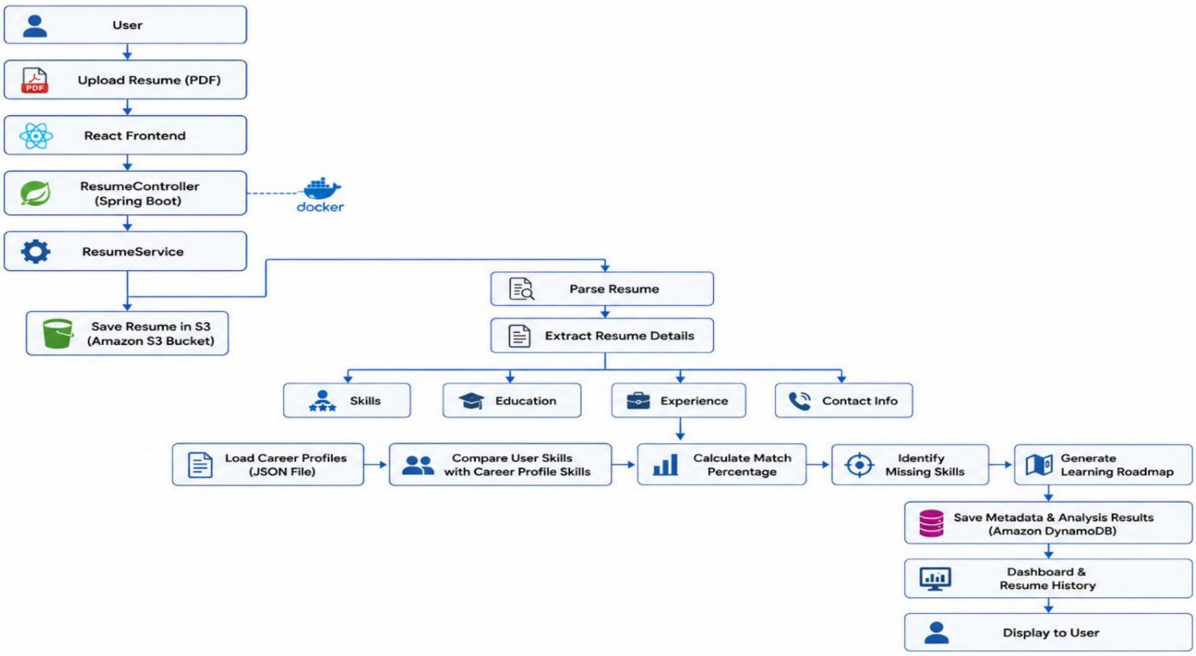

Figure 4.4 Resume Processing Workflow

4.5 Database Layer – Amazon DynamoDB

Amazon DynamoDB is used to store resume metadata and analysis information. It stores details such as the resume name, upload date, resume score, recommended career path, and resume history. DynamoDB provides fast data access and supports efficient storage of application data.

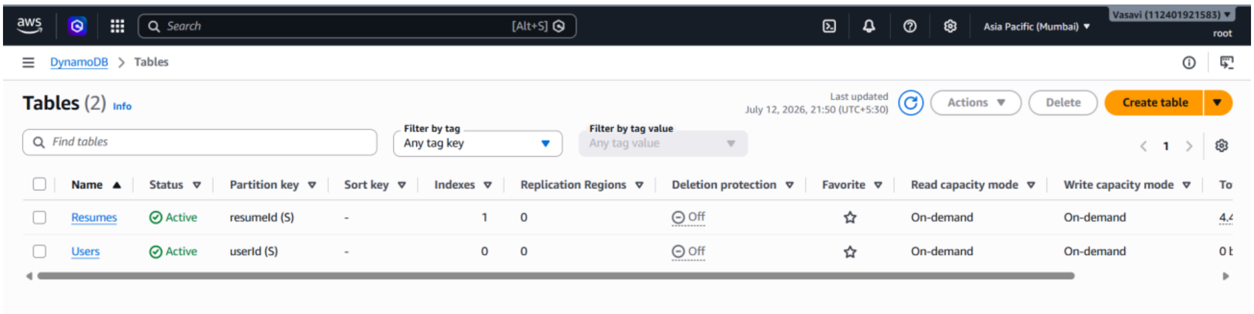

Figure 4.5.1 DynamoDB Table

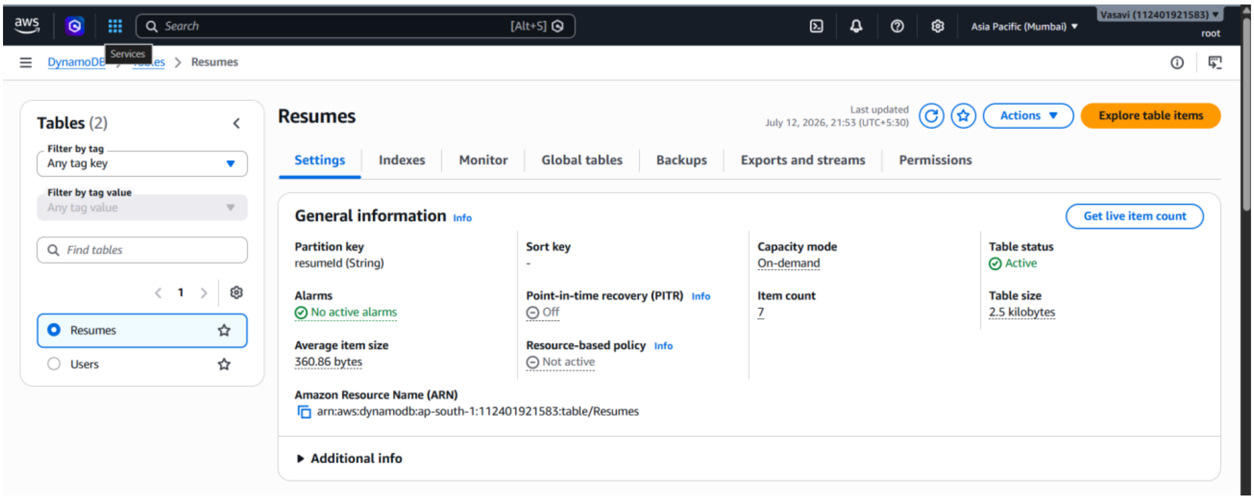

Figure 4.5.2 DynamoDB Table for Resumes

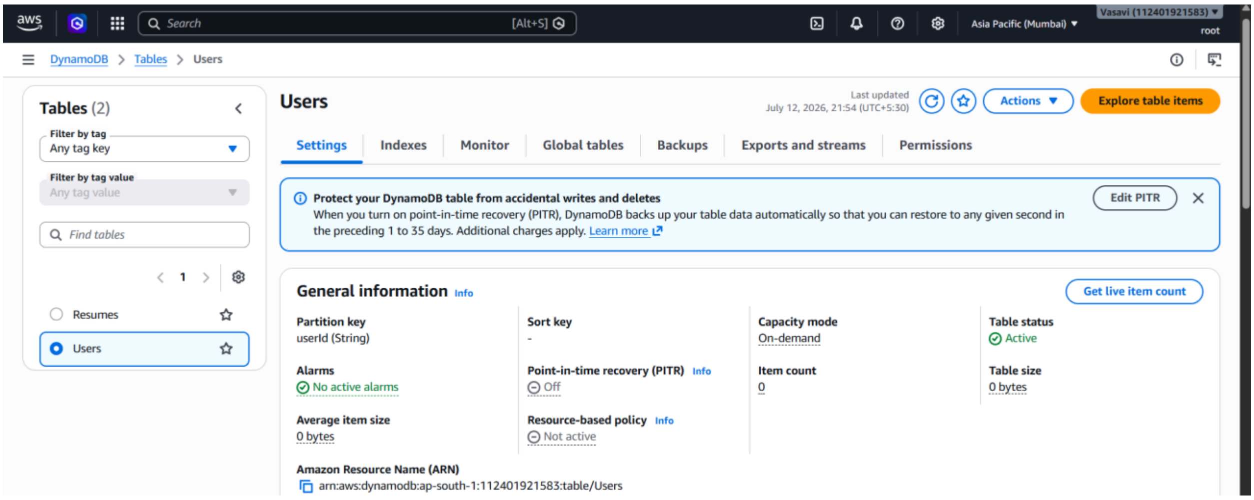

Figure 4.5.3 DynamoDB Table for Users

4.6 Security Layer – AWS IAM

AWS Identity and Access Management (IAM) is used to provide secure access between the backend application and AWS services. An IAM user with the required permissions is created to allow the backend to access Amazon S3 and DynamoDB securely. IAM helps protect AWS resources by controlling user permissions and access.

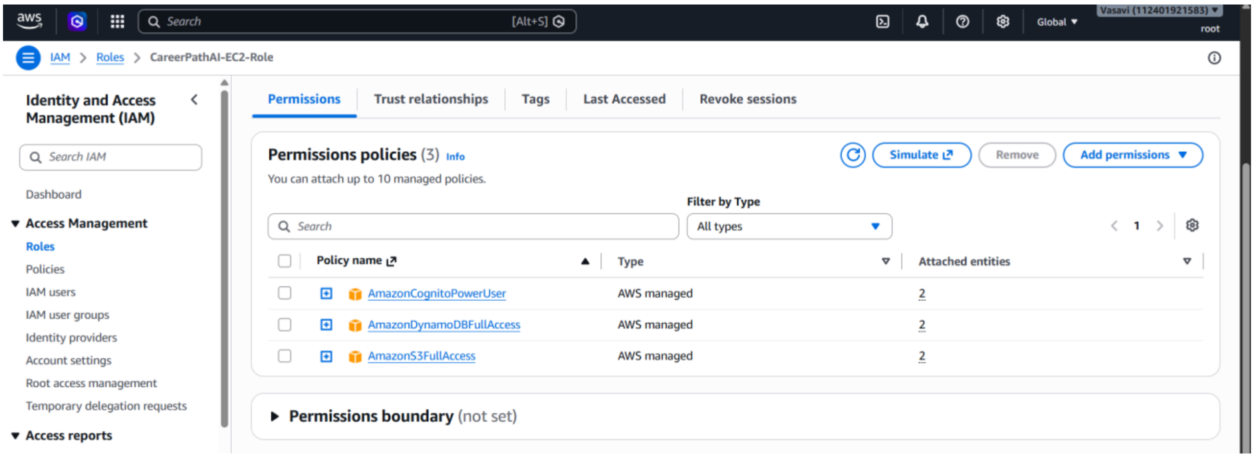

Figure 4.6 IAM User

4.7 Deployment Architecture – Amazon EC2 and Docker

The Spring Boot backend application is deployed on an Amazon EC2 instance. Docker is used to package the backend application into a container before deployment. Running the application inside a Docker container makes deployment easier and ensures that the application works consistently in different environments. Amazon EC2 provides a reliable cloud server for hosting the backend application.

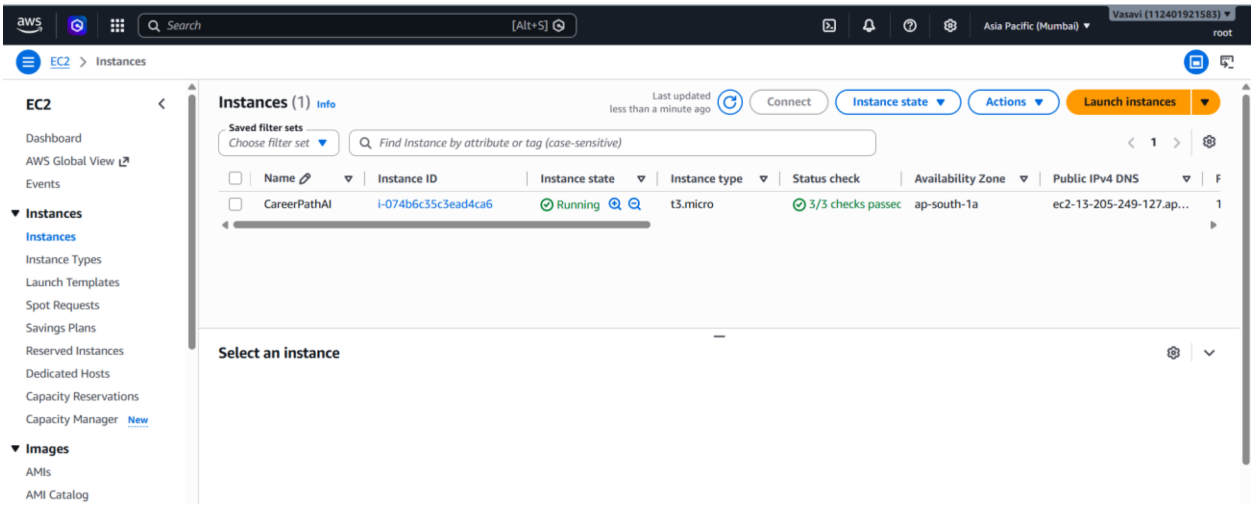

Figure 4.7 Amazon EC2 Instance

4.8 Network Configuration – Elastic IP

An Elastic IP is associated with the Amazon EC2 instance to provide a fixed public IP address. Normally, the public IP of an EC2 instance changes whenever it is restarted. By using an Elastic IP, the backend application always remains available through the same IP address, allowing the React frontend to communicate with the backend without changing the API URL.

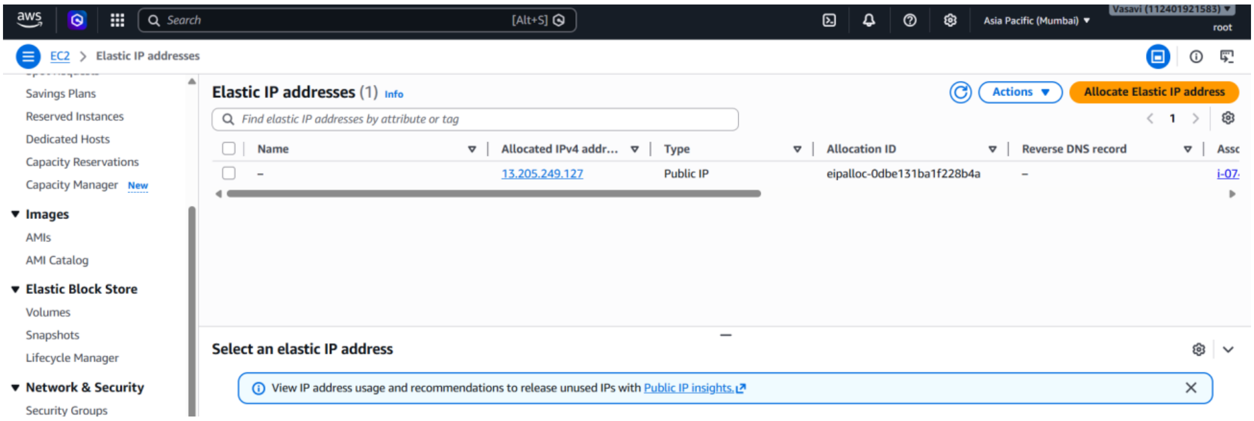

Figure 4.8 Elastic IP

4.9 Application Workflow

The application workflow starts when the user registers and logs in using Amazon Cognito. After successful authentication, the user uploads a resume in PDF format. The resume is stored in Amazon S3, and the backend extracts the required skills. These skills are compared with predefined career profiles stored in a JSON file to identify the best matching career and missing skills. A personalized learning roadmap is then generated, and the analysis results are stored in Amazon DynamoDB. Finally, the user can view all the results through the dashboard.

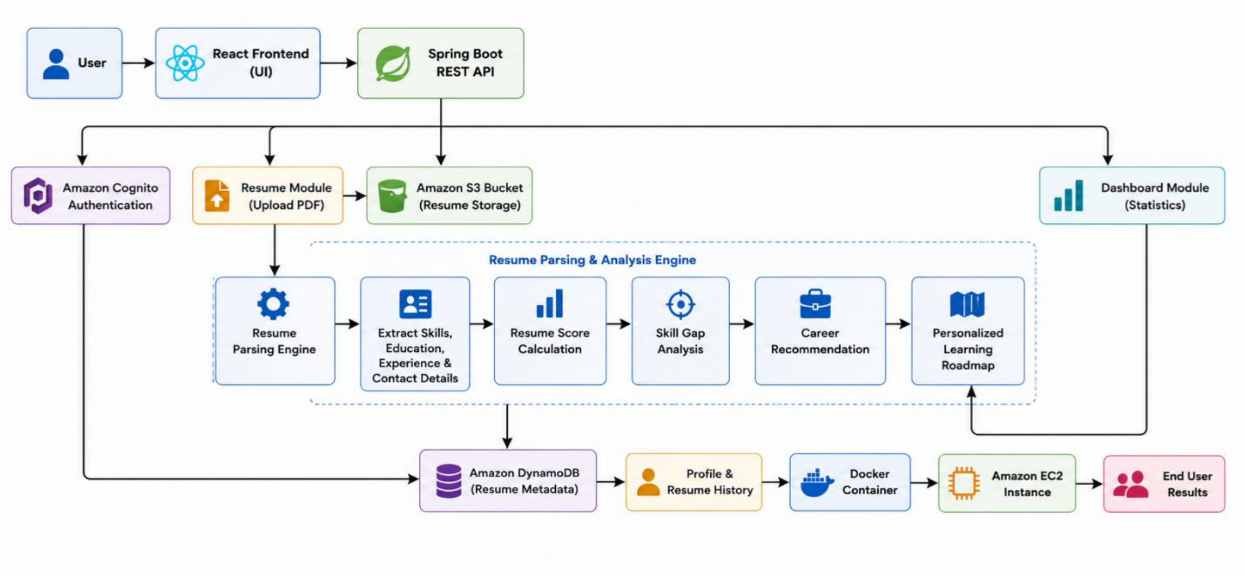

Figure 4.9 Application Workflow

5. IMPLEMENTATION

This chapter explains how each module of the CareerPath AI application was developed and how they work together. The implementation is divided into different modules, including user authentication, resume upload, resume processing, career recommendation, dashboard, and deployment. Each module performs a specific task to provide a secure and user-friendly experience.

5.1 User Authentication Module

The User Authentication module is responsible for managing user registration and login. Amazon Cognito is used to authenticate users securely. New users can create an account using their email address and password. After registration, a verification code is sent to the user's email. Once the email is verified, the user can log in to the application.

After successful login, Amazon Cognito generates authentication tokens. These tokens are sent with every request and validated by Spring Security before allowing access to protected APIs.

Functions

- User Registration
- Email Verification
- User Login
- Forgot Password
- Reset Password
- JWT Token Authentication

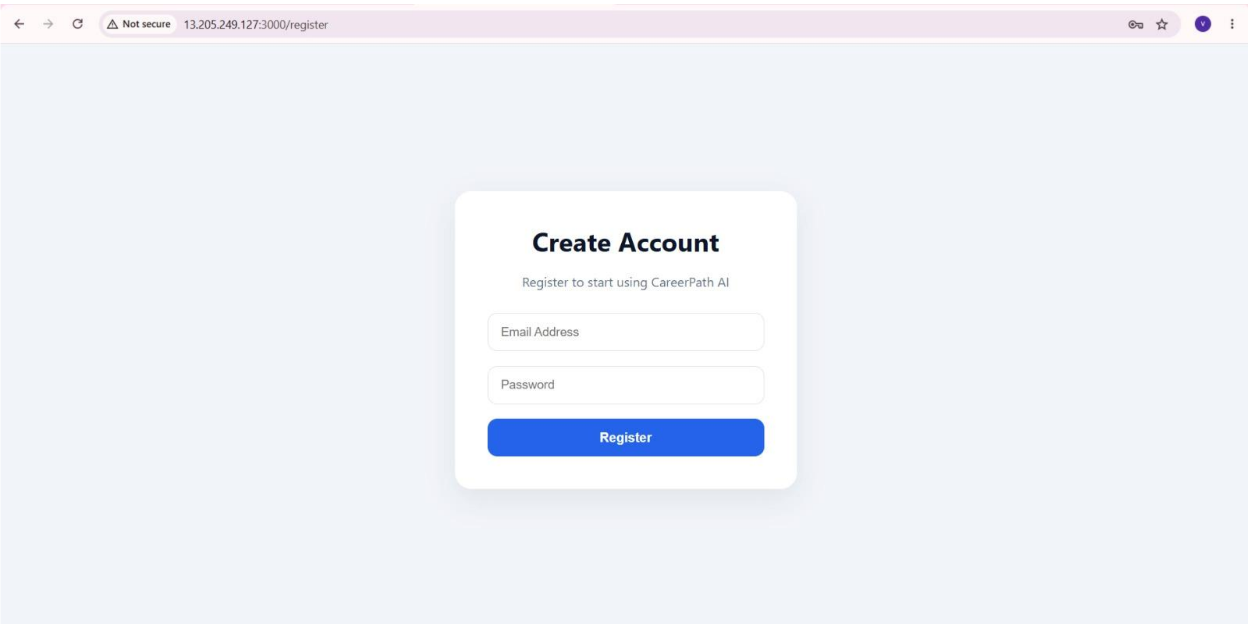

Figure 5.1 User Authentication Module

5.2 Resume Upload Module

The Resume Upload module allows users to upload their resumes in PDF format. The React frontend sends the selected file to the Spring Boot backend through a REST API. The backend validates the file and uploads it to Amazon S3. Resume information is also stored in Amazon DynamoDB for future reference.

This module ensures that resumes are uploaded securely and are available for further processing.

Functions

- Select Resume
- Upload PDF
- Store Resume in Amazon S3
- Save Resume Details

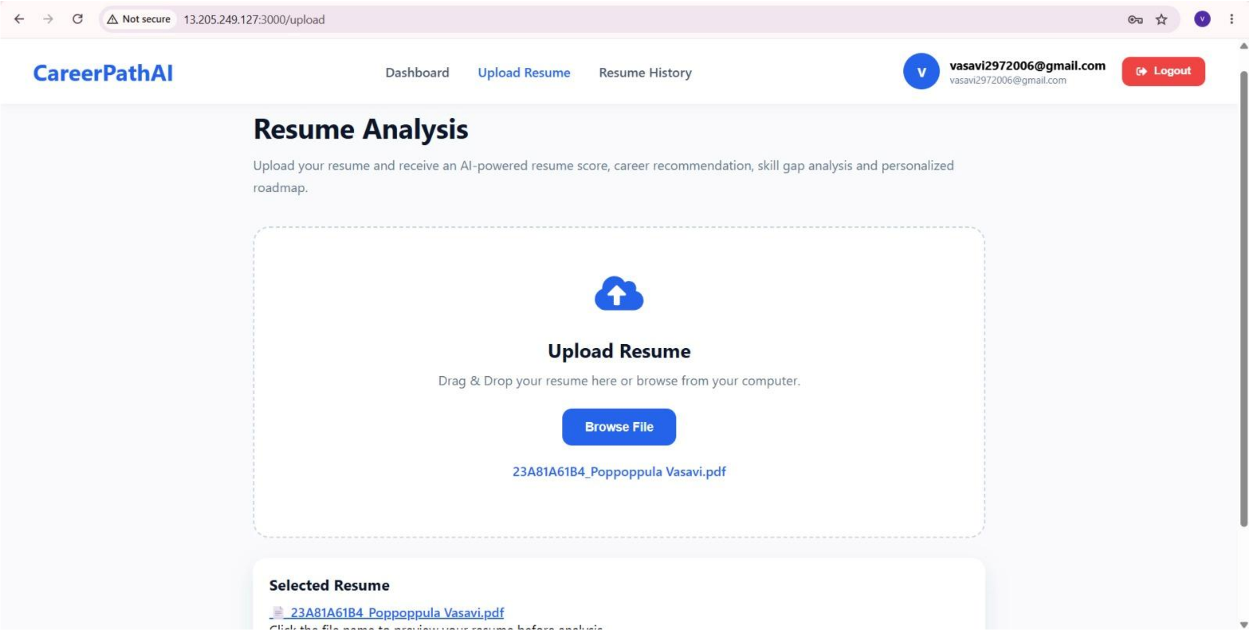

Figure 5.2 Resume Upload Module

5.3 Resume Analysis Module

The Resume Analysis module processes the uploaded resume. The backend extracts information such as skills, education, projects, and experience. The extracted skills are compared with **predefined career profiles stored in a JSON file**.

The application calculates the career match percentage, identifies missing skills, and selects the most suitable career profile. Based on the missing skills, a personalized learning roadmap is generated for the user.

Functions

- Resume Parsing

- Skill Extraction
- JSON Profile Matching
- Career Recommendation
- Missing Skills Identification
- Learning Roadmap Generation

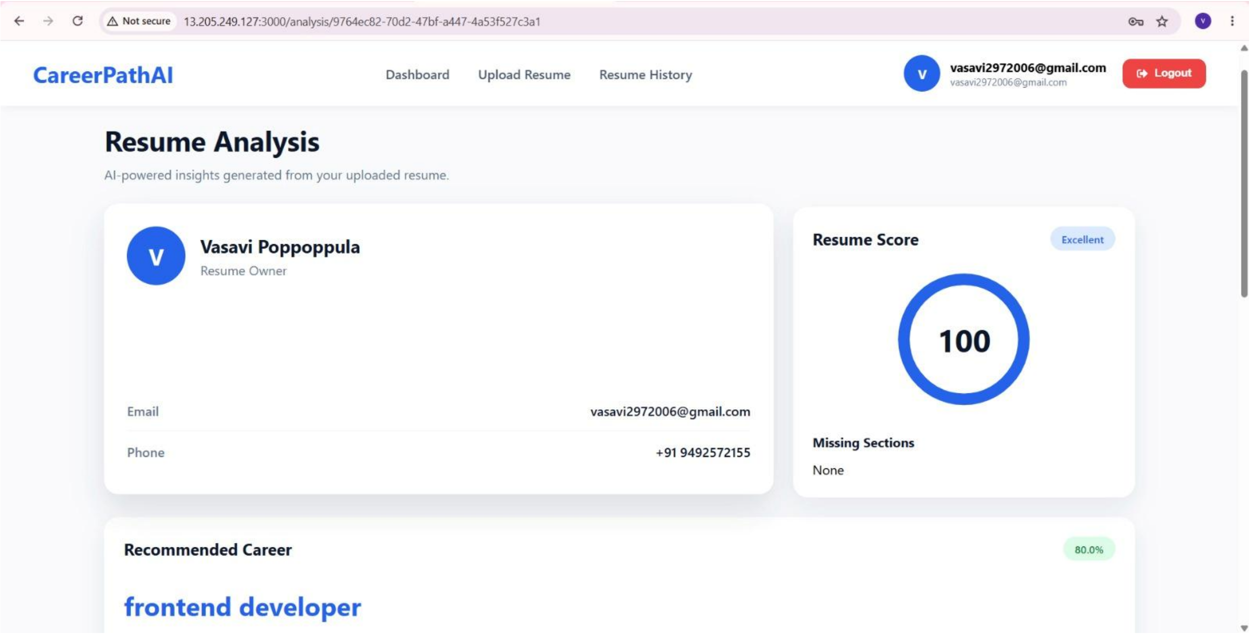

Figure 5.3 Resume Analysis Module

5.4 Dashboard Module

The Dashboard module displays the analysis results in an easy-to-understand format. Users can view their resume score, career recommendation, missing skills, learning roadmap, and uploaded resume history.

The dashboard retrieves the required data from the backend and displays it through the React frontend.

Functions

- Resume Score
- Career Recommendation
- Missing Skills
- Learning Roadmap
- Resume History

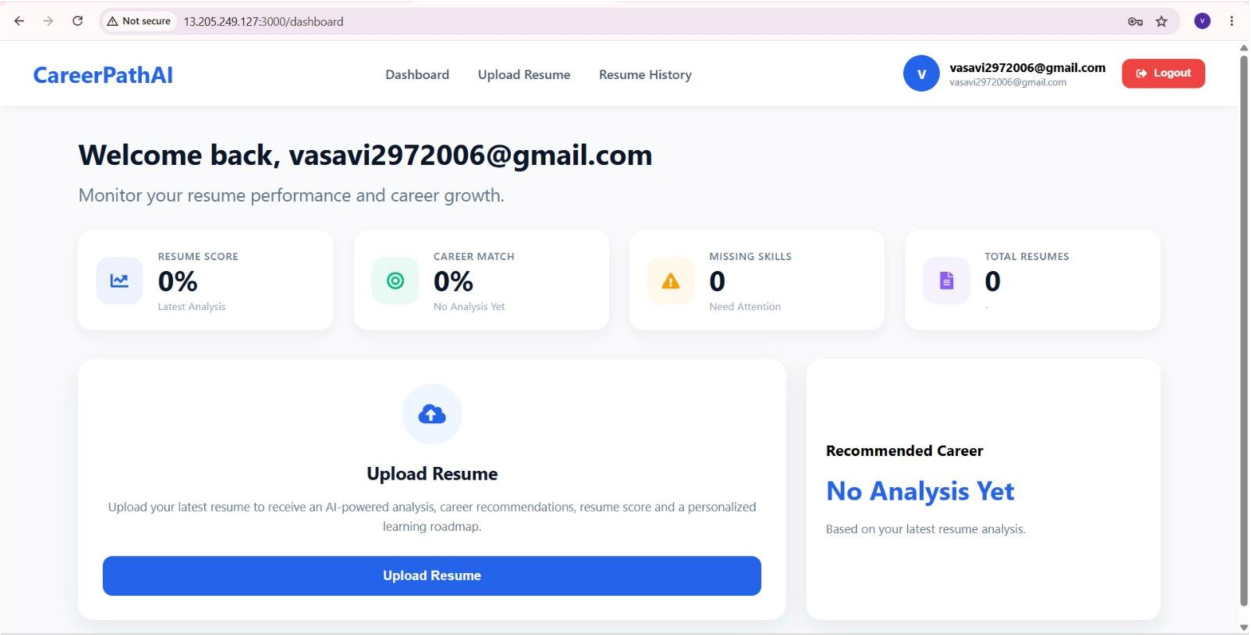

Figure 5.4 Dashboard Module

5.5 Deployment Module

After development, the backend application was containerized using Docker and deployed on an Amazon EC2 instance. An Elastic IP was associated with the EC2 instance to provide a fixed public IP address. The frontend communicates with the backend through this fixed endpoint.

Git and GitHub were used to manage the source code throughout the development process.

Functions

- Docker Container
- EC2 Deployment
- Elastic IP Configuration
- GitHub Repository Management

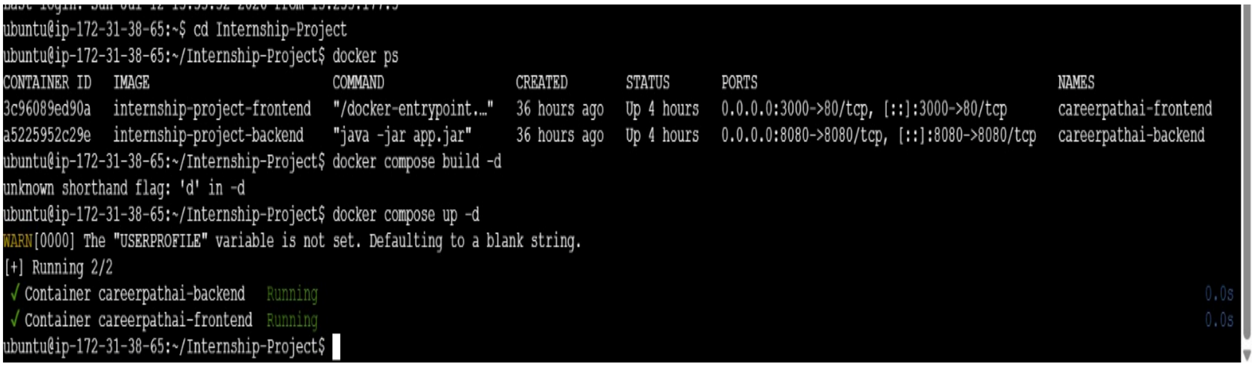

Figure 5.5.1 EC2 Deployment Module

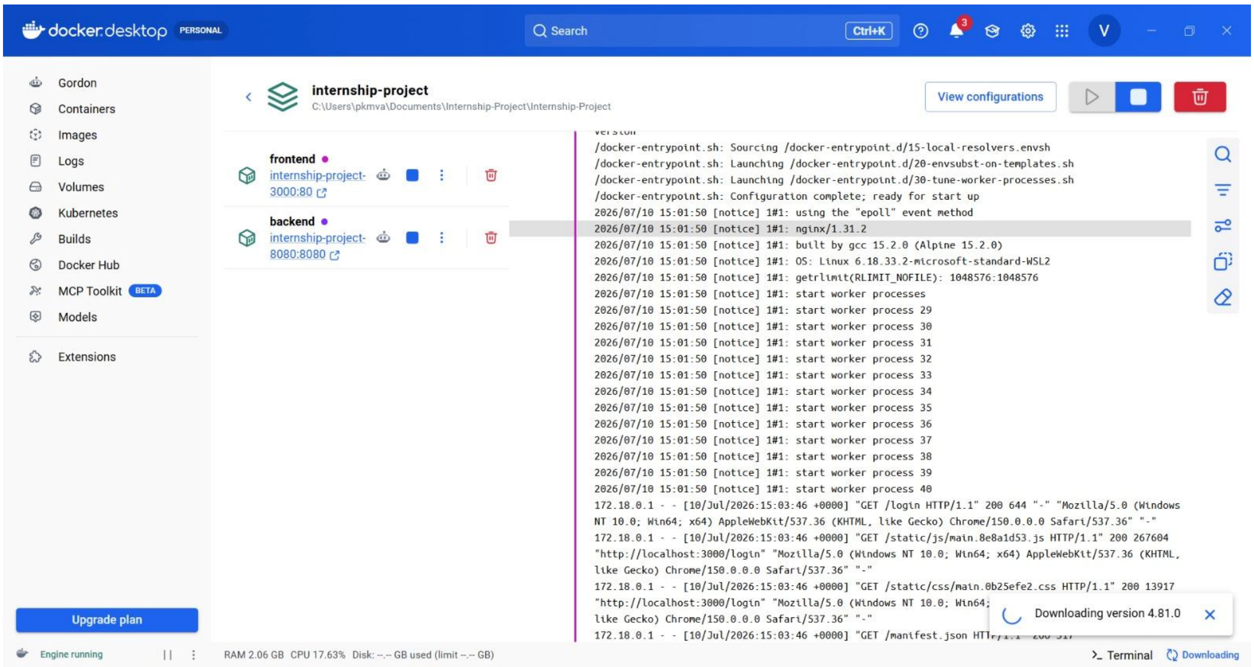

Figure 5.5.1 Docker Deployment Module

6. TESTING AND RESULTS

Testing is an important phase in software development. It ensures that every module of the application works correctly and produces the expected results. In CareerPath AI, all major features were tested individually before integrating the complete application. The testing process included user authentication, resume upload, resume analysis, dashboard functionality, and deployment. The results confirmed that the application works correctly and meets the project requirements.

6.1 Authentication Testing

The authentication module was tested to verify user registration, email verification, login, forgot password, and reset password functionalities. Amazon Cognito successfully authenticated users and generated JWT tokens for secure access to the application.

Table 6.1 Authentication Test Cases

Test Case	Expected Result	Status
User Registration	User account created successfully	Pass
Email Verification	Email verified successfully	Pass
User Login	User logged in successfully	Pass
Forgot Password	Reset code sent to registered email	Pass
Reset Password	Password updated successfully	Pass

6.2 Resume Upload Testing

The resume upload module was tested by uploading valid PDF resumes. The application successfully stored the uploaded resumes in Amazon S3 and saved the corresponding metadata in Amazon DynamoDB.

Table 6.2 Resume Upload Test Cases

Test Case	Expected Result	Status
Upload Valid PDF	Resume uploaded successfully	Pass
Upload Invalid File	Error message displayed	Pass
Store Resume in Amazon S3	Resume stored successfully	Pass
Save Resume Metadata	Data saved in DynamoDB	Pass

6.3 Resume Analysis Testing

The resume analysis module was tested using different sample resumes. The application extracted user skills from the resume, compared them with predefined career profiles stored in a JSON file, identified missing skills, calculated the career match percentage, and generated a personalized learning roadmap.

Table 6.3 Resume Analysis Test Cases

Test Case	Expected Result	Status
Resume Parsing	Resume parsed successfully	Pass
Skill Extraction	Skills extracted correctly	Pass
Career Profile Matching	Correct career profile selected	Pass
Missing Skills Identification	Missing skills displayed	Pass
Learning Roadmap Generation	Roadmap generated successfully	Pass

6.4 Dashboard Testing

The dashboard module was tested to verify that all analysis results were displayed correctly. The dashboard successfully displayed resume score, career recommendation, missing skills, learning roadmap, and resume history.

Table 6.4 Dashboard Test Cases

Test Case	Expected Result	Status
Display Resume Score	Score displayed correctly	Pass
Display Career Recommendation	Recommendation displayed	Pass
Display Missing Skills	Missing skills displayed	Pass
Display Learning Roadmap	Roadmap displayed correctly	Pass
Display Resume History	Resume history displayed	Pass

6.5 Deployment Testing

The backend application was deployed on an Amazon EC2 instance using Docker. The deployment was tested by accessing the backend through the Elastic IP address. All REST APIs responded correctly, and the frontend communicated successfully with the backend.

Table 6.5 Deployment Test Cases

Test Case	Expected Result	Status
EC2 Instance Running	Backend server available	Pass
Docker Container Running	Application running successfully	Pass
Backend API Access	APIs responded correctly	Pass
Elastic IP Connectivity	Backend accessible through Elastic IP	Pass

6.6 Overall Results

The testing results show that all modules of the CareerPath AI application work successfully. User authentication, resume upload, resume analysis, dashboard, and deployment modules were tested and produced the expected results. The application securely stores user data and resumes using AWS services and provides career recommendations by comparing user skills with predefined career profiles stored in a JSON file.

The project successfully achieved its objectives by integrating React, Spring Boot, Docker, and AWS cloud services into a secure and user-friendly career guidance platform.

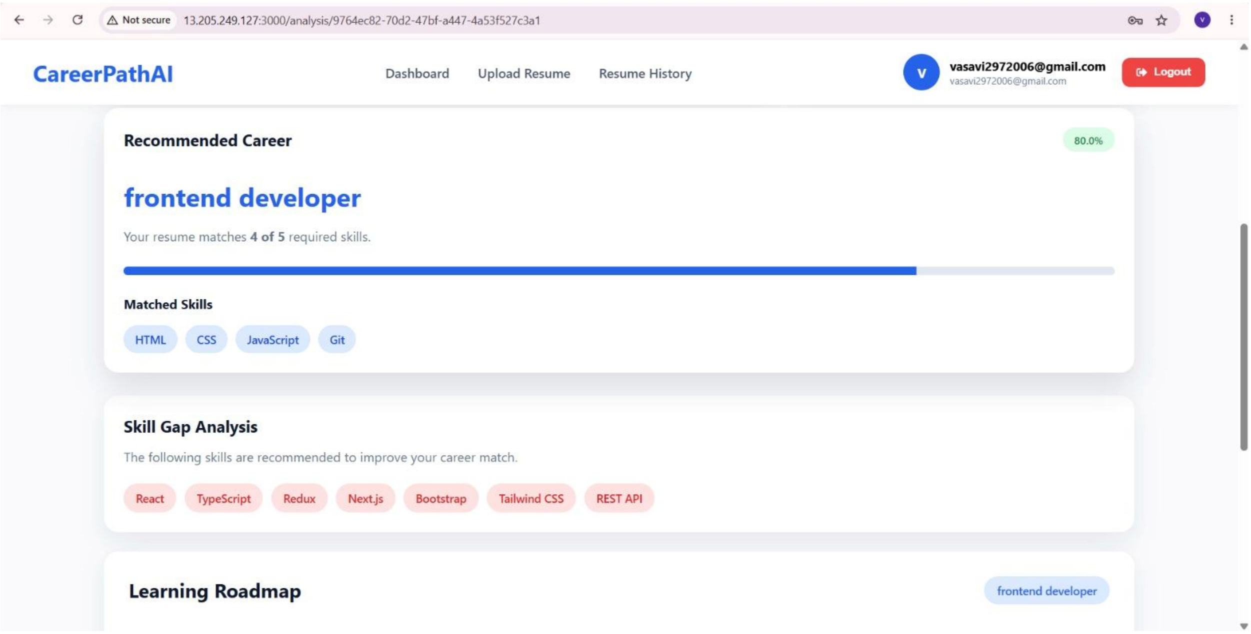

Figure 6.6 : Final Output

7. CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The CareerPath AI project was successfully designed and developed as a secure web application to help students and job seekers analyze their resumes and receive career guidance. The application allows users to register and log in securely using Amazon Cognito, upload resumes, and view personalized career recommendations.

The uploaded resumes are stored in Amazon S3, while resume metadata and analysis results are stored in Amazon DynamoDB. The application extracts skills from the uploaded resume and compares them with predefined career profiles stored in a JSON file. Based on the matching skills, the system identifies missing skills, recommends a suitable career path, and generates a personalized learning roadmap.

The backend application was deployed on Amazon EC2 using Docker, and an Elastic IP was associated with the EC2 instance to provide a fixed public IP address. The project successfully integrates React, Spring Boot, Docker, and AWS services to build a secure and user-friendly full-stack application.

Overall, the project achieved all its objectives and provided valuable practical experience in full-stack development, cloud computing, secure authentication, and application deployment.

7.2 Future Scope

CareerPath AI can be enhanced by adding more features in the future. Some possible improvements are:

- Add more career profiles for better recommendations.
- Support additional resume formats such as DOCX.
- Improve skill matching with more detailed career requirements.
- Add interview preparation and practice questions.
- Integrate job recommendation features.
- Provide resume improvement suggestions.
- Develop a mobile application.
- Add analytics and progress tracking for users.
- Add Built-in AI models to analyses Resumes Accurately.

These improvements can make the application more useful for students and job seekers.